

Name: SANTHOSH M

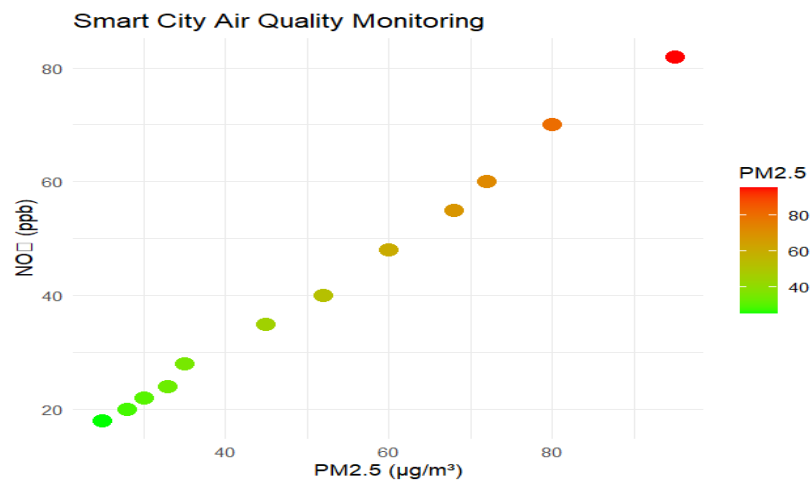
Reg No: 192324014

Scenario :1 Smart City Air Quality Monitoring.

Code:

```
> library(ggplot2)
>
> air_data <- read.csv("air_quality_data(1).csv")
> ggplot(air_data,
+   aes(x = PM2.5,
+     y = NO2,
+     color = PM2.5)) +
+   geom_point(size = 4) +
+   scale_color_gradient(low = "green", high = "red") +
+   labs(
+     title = "Smart City Air Quality Monitoring",
+     x = "PM2.5 (µg/m³)",
+     y = "NO₂ (ppb)",
+     color = "PM2.5"
+   ) +
+   theme_minimal()
```

Output:

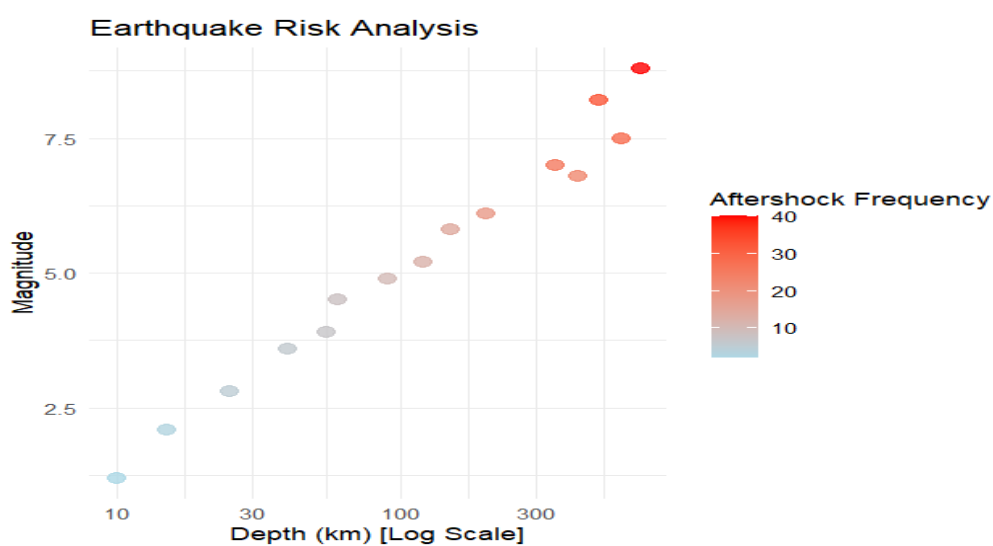


Scenario : 2 Earthquake Risk Analysis

Code:

```
> library(ggplot2)
>
> earthquake_data <- read.csv("earthquake_risk_analysis.csv")
>
> head(earthquake_data)
  Earthquake_ID Magnitude Depth Latitude Longitude Aftershock_Frequency
1      EQ001      1.2    10   11.5    76.9         2
2      EQ002      2.8    25   12.1    77.3         5
3      EQ003      4.5    60   13.8    78.1         8
4      EQ004      5.2   120   15.2    79.4        12
5      EQ005      6.1   200   16.5    80.1        18
6      EQ006      7.0   350   18.2    81.3        25
>
> ggplot(earthquake_data,
+   aes(x = Depth,
+       y = Magnitude,
+       color = Aftershock_Frequency)) +
+   geom_point(size = 3, alpha = 0.8) +
+   scale_color_gradient(low = "lightblue", high = "red") +
+   scale_x_log10() +
+   labs(
+     title = "Earthquake Risk Analysis",
+     x = "Depth (km) [Log Scale]",
+     y = "Magnitude",
+     color = "Aftershock Frequency"
+   ) +
+   theme_minimal()
```

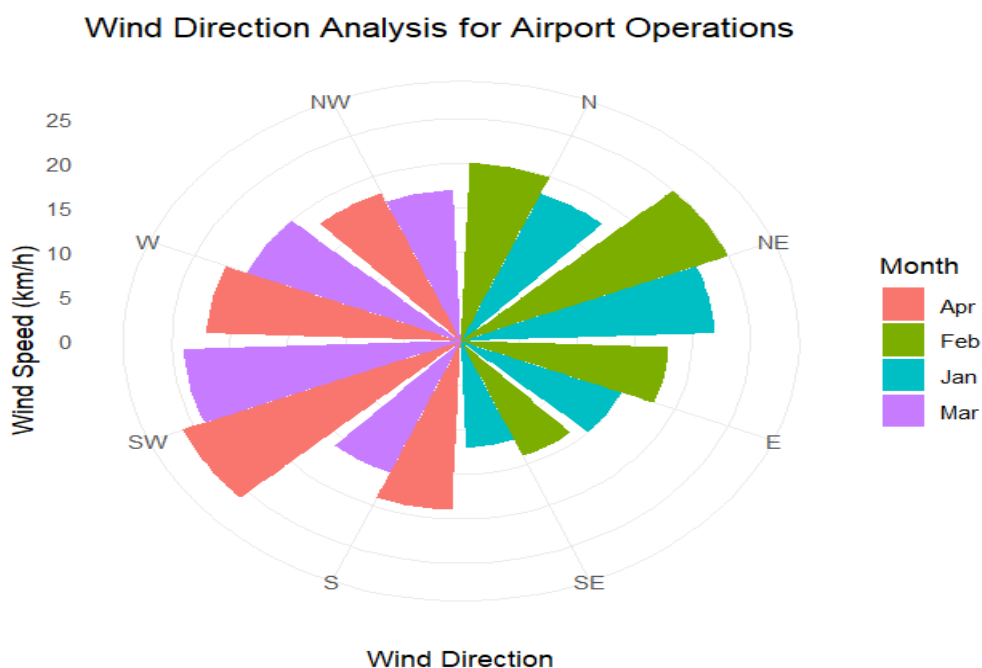
Output:



Scenario : 3 Wind Direction Analysis for Airport OperationsCode:

```
> library(ggplot2)
>
> wind_data <- read.csv("wind_direction_analysis.csv")
> wind_data$Wind_Direction <- factor(
+   wind_data$Wind_Direction,
+   levels = c("N", "NE", "E", "SE", "S", "SW", "W", "NW"))
+ )
>
> ggplot(wind_data,
+   aes(x = Wind_Direction,
+     y = Wind_Speed,
+     fill = Month)) +
+   geom_bar(stat = "identity", position = "dodge") +
+   coord_polar() +
+   labs(
+     title = "Wind Direction Analysis for Airport Operations",
+     x = "Wind Direction",
+     y = "Wind Speed (km/h)"
+   ) +
+   theme_minimal()
```

Output:

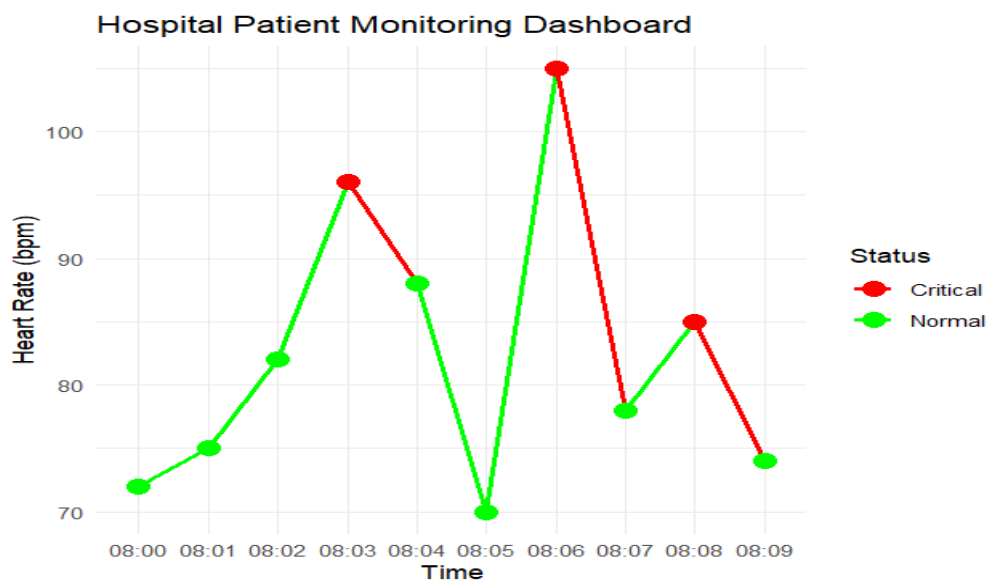


Scenario : 4 Hospital Patient Monitoring Dashboard

Code:

```
> library(ggplot2)
>
> patient_data <- read.csv("hospital_patient_monitoring.csv")
> patient_data$Status <- ifelse(patient_data$Heart_Rate > 100 |
+                               patient_data$Oxygen_Saturation < 94 |
+                               patient_data$Temperature > 38,
+                               "Critical", "Normal")
>
> ggplot(patient_data,
+       aes(x = Time,
+           y = Heart_Rate,
+           color = Status,
+           group = 1)) +
+   geom_line(linewidth = 1) +
+   geom_point(size = 4) +
+   scale_color_manual(values = c("Normal" = "green",
+                                   "Critical" = "red")) +
+   labs(
+     title = "Hospital Patient Monitoring Dashboard",
+     x = "Time",
+     y = "Heart Rate (bpm)"
+   ) +
+   theme_minimal()
```

Output:



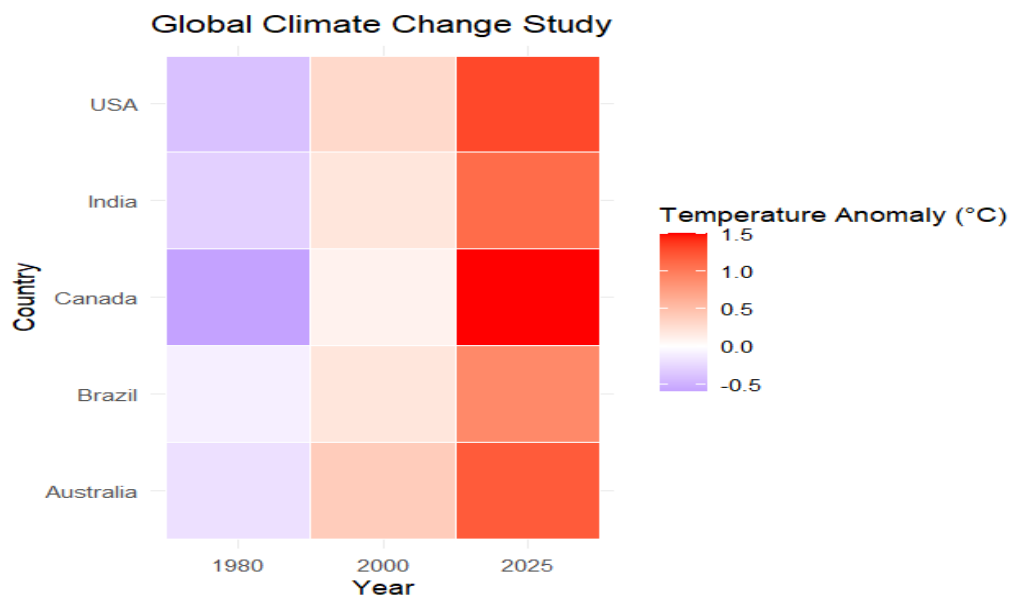
Scenario : 5 Global Climate Change Study

Code:

```
> library(ggplot2)

> climate_data <- read.csv("global_climate_change.csv")
> ggplot(climate_data,
+   aes(x = factor(Year),
+     y = Country,
+     fill = Temperature_Anomaly)) +
+   geom_tile(color = "white") +
+   scale_fill_gradient2(
+     low = "blue",
+     mid = "white",
+     high = "red",
+     midpoint = 0
+   ) +
+   labs(
+     title = "Global Climate Change Study",
+     x = "Year",
+     y = "Country",
+     fill = "Temperature Anomaly (°C)"
+   ) +
+   theme_minimal()
```

Output:

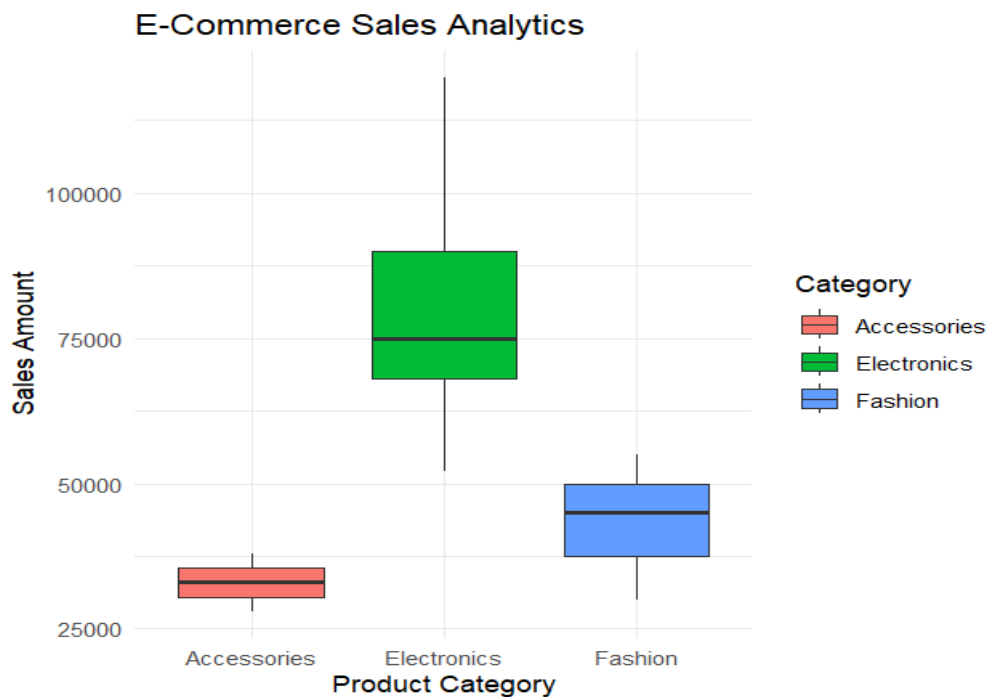


Scenario : 6 E-Commerce Sales Analytics

Code:

```
> library(ggplot2)
>
> sales_data <- read.csv("ecommerce_sales_analytics.csv")
>
> ggplot(sales_data,
+   aes(x = Category,
+     y = Sales,
+     fill = Category)) +
+   geom_boxplot() +
+   labs(
+     title = "E-Commerce Sales Analytics",
+     x = "Product Category",
+     y = "Sales Amount"
+   ) +
+   theme_minimal()
>
>
>
> library(ggplot2)
```

Output:

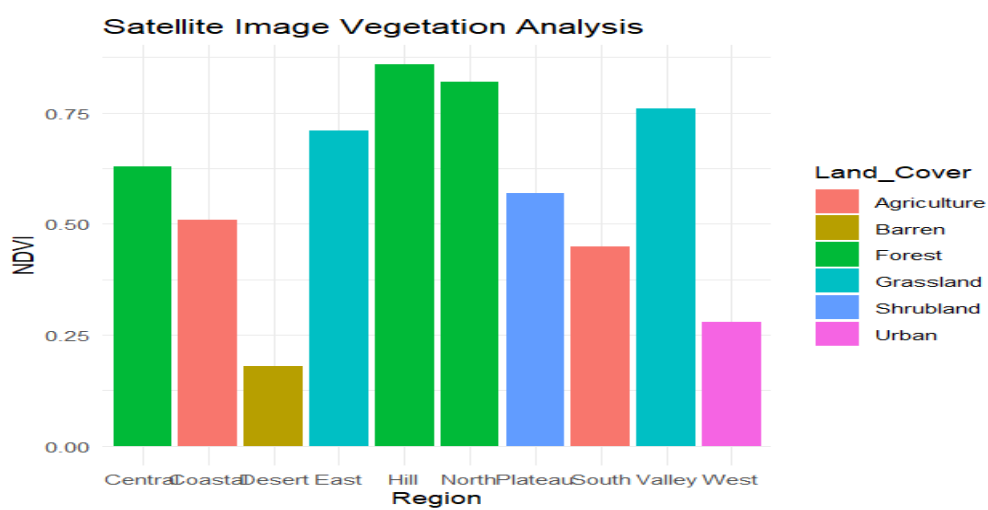


Scenario : 7 Satellite Image Vegetation Analysis

Code:

```
> library(ggplot2)
>
> vegetation_data <- read.csv("satellite_vegetation_analysis.csv")
>
> ggplot(vegetation_data,
+   aes(x = Region,
+     y = NDVI,
+     fill = Land_Cover)) +
+   geom_col() +
+   labs(
+     title = "Satellite Image Vegetation Analysis",
+     x = "Region",
+     y = "NDVI"
+   ) +
+   theme_minimal()
>
>
>
> library(ggplot2)
```

Output:



Scenario : 8 Cryptocurrency Market Volatility

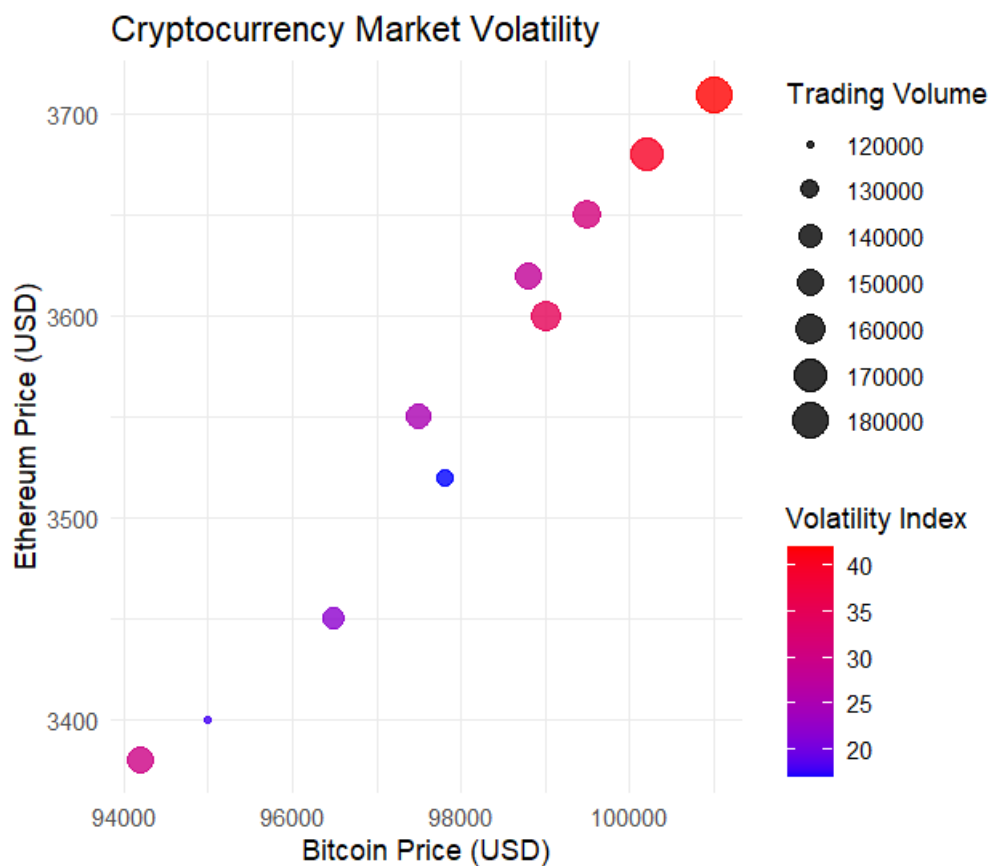
Code:

```

> library(ggplot2)
>
> crypto_data <- read.csv("cryptocurrency_market_volatility.csv")
>
> ggplot(crypto_data,
+   aes(x = Bitcoin,
+     y = Ethereum,
+     size = Trading_Volume,
+     color = Volatility_Index)) +
+   geom_point(alpha = 0.8) +
+   scale_color_gradient(low = "blue", high = "red") +
+   labs(
+     title = "Cryptocurrency Market Volatility",
+     x = "Bitcoin Price (USD)",
+     y = "Ethereum Price (USD)",
+     color = "Volatility Index",
+     size = "Trading Volume"
+   ) +
+   theme_minimal()

```

Output:



Scenario : 9 Marathon Runner Performance Analysis

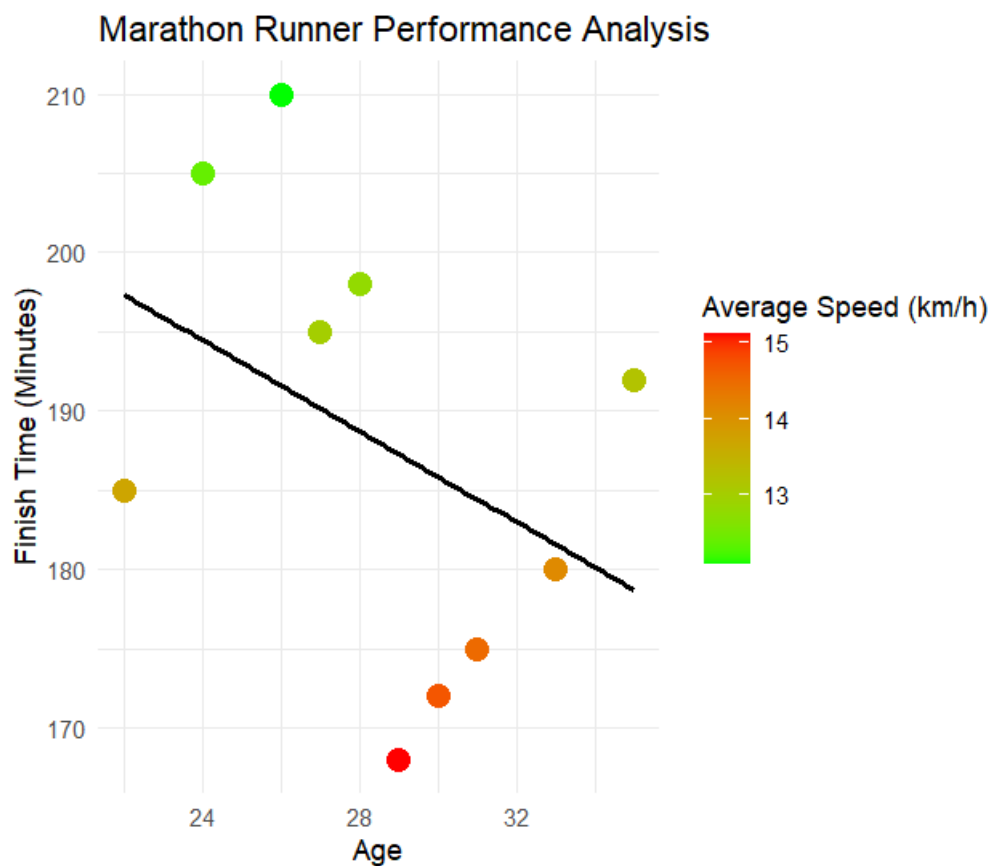
Code:


```

> library(ggplot2)
>
> runner_data <- read.csv("marathon_runner_performance.csv")
>
> ggplot(runner_data,
+   aes(x = Age,
+     y = Finish_Time_Minutes,
+     color = Average_Speed_kmph)) +
+   geom_point(size = 4) +
+   geom_smooth(method = "lm", se = FALSE, color = "black") +
+   scale_color_gradient(low = "green", high = "red") +
+   labs(
+     title = "Marathon Runner Performance Analysis",
+     x = "Age",
+     y = "Finish Time (Minutes)",
+     color = "Average Speed (km/h)"
+   ) +
+   theme_minimal()
`geom_smooth()` using formula = 'y ~ x'

```

Output:



Scenario : 10 COVID-19 Vaccination Campaign Dashboard

Code:

```
library(ggplot2)

iris_data <- read.csv("iris_classification.csv")

head(iris_data)

ggplot(iris_data,
       aes(x = Sepal_Length,
           y = Petal_Length,
           color = Species,
           shape = Species,
           size = Sepal_Width)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Iris Flower Classification",
    x = "Sepal Length",
    y = "Petal Length",
    size = "Sepal Width"
  ) +
  theme_minimal()
```

Output:

